# Serial Communications with SPI

ECE 2564, Embedded Systems

Fall 2025

# Overview

Serial Peripheral Interface (SPI)

- Serial, synchronous, full-duplex
- Developed by Motorola in the 1980s and has become a de facto standard
- Usually used at the board level
- Often called a 4-wire serial bus
  - Better described as a $(3 + n)$-wire bus, where $n$ is the number of devices on the bus

# SPI Example 1

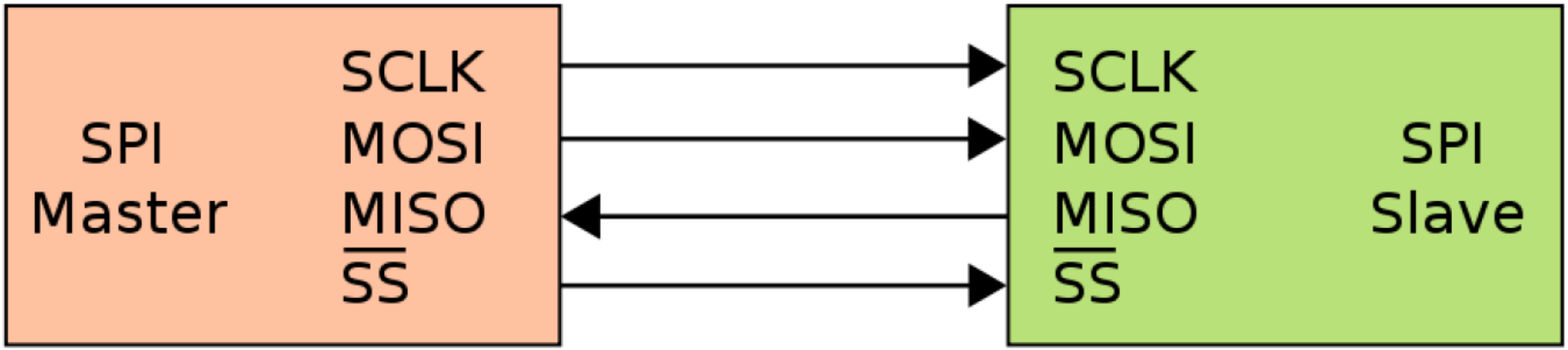



Diagram courtesy of Wikipedia

- SCLK: Serial Clock
- MOSI: Master Out, Slave In (aka SIMO)
  - Similar to Tx in UART (for the master)
- MISO: Master In, Slave Out (aka SOMI)
  - Similar to Rx in UART (for the master)
- $\overline{\text{SS}}$: Slave Select (active low)

# SPI Interface

- SCLK (*Serial Clock*): Generated by the Master to synchronize the transfers
- MOSI (*Master Out Slave In*): Carries data from the Master to the Slave
- MISO (*Master In Slave Out*): Carries data from the Slave to the Master
- SS (*Slave Select*): Generated by the Master to *enable* a particular peripheral device
- In SPI, there is no start or stop condition
  - The initiator/master lowers SS (Slave/Chip select) to tell the follower/slave that communication is starting
  - Bringing the signal back to high tells the follower/slave that communication is over

# SPI Arrangment – In Concept

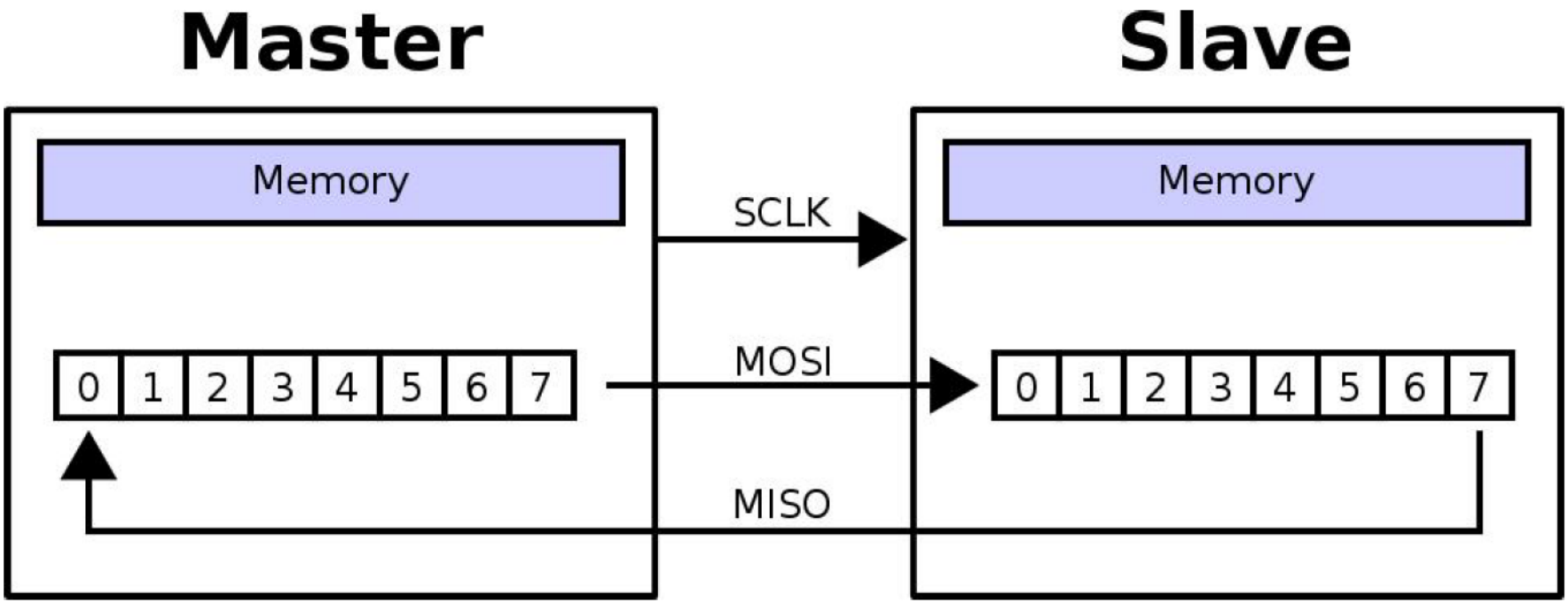
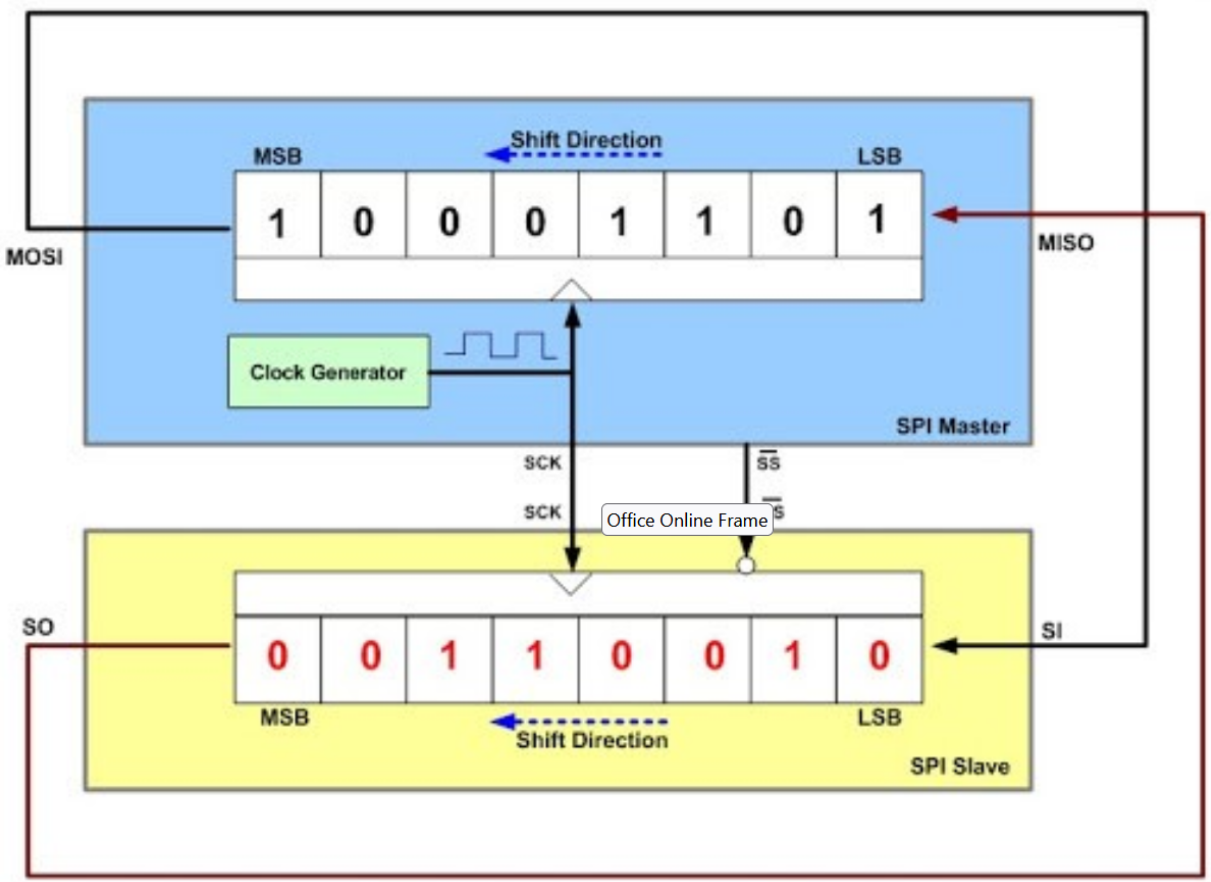



Diagram courtesy of Wikipedia

- Use two shift registers to form an *inter-chip circular buffer*
  - MSB from master is shifted into LSB of slave
  - MSB from slave is shifted into LSB of master
- $\overline{SS}$ signal is not shown in this diagram

# SPI Master and Slave Interconnection

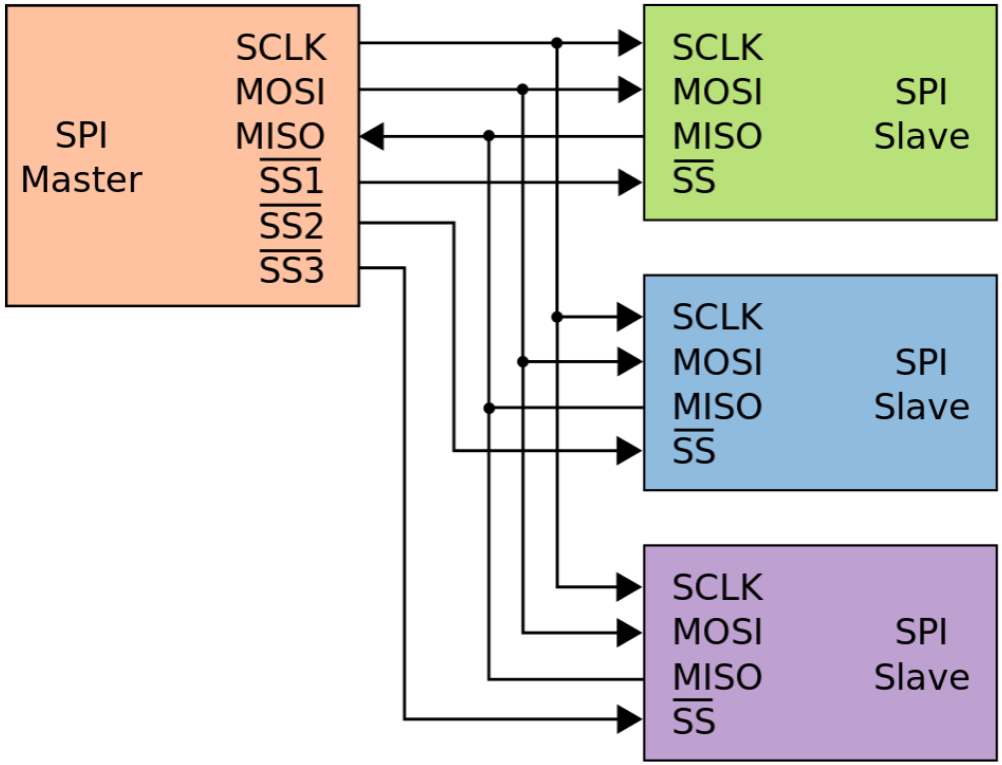



What are the values of the registers after one cycle?

# SPI Master and Slave Interconnection

- For SPI sending data automatically means receiving data, too
- Any communication has three steps:
  - Copy from Transmit buffer to shift register
  - Shift eight times
  - Copy from shift register to Receive buffer
- Unlike $I^2C$, SPI does not support multiple masters
- There are two ways for a single initiator to connect to multiple followers:
  - Direct configuation
  - Daisy-chain configuration

# SPI Direct Configuration for Multiple Peripherals

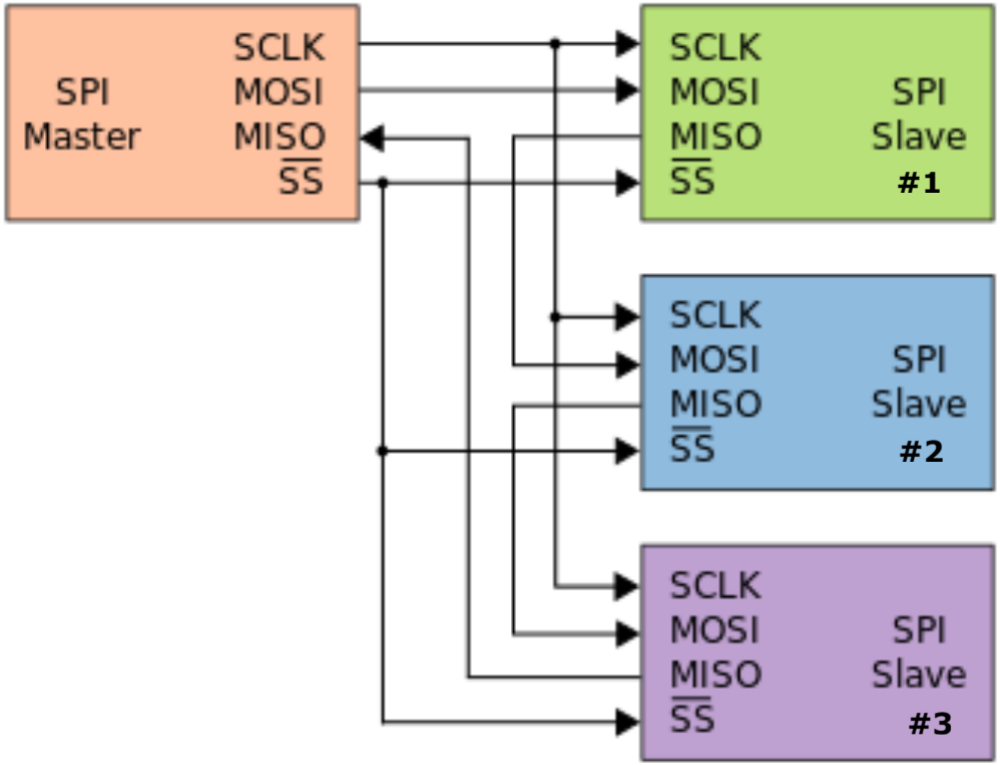



- Shared clock
- What about MOSI and MISO?

# SPI Direct Configuration: Arrangement and Timing

- Only one device can be the master at any given time in any given system
- Serveral slave devices can be present
  - Only one is in communication with the master at any given time
  - $\overline{SS}$ signal must be asserted for that specific slave device
- During one "group" of clock cycles the master shift data out to the specific slave
- The master received data that is being shifted fro the same slave device

# SPI Daisy Configuration for Multiple Peripherals




- Shared clock
- Data lines are connected in a sequence or chain

# SPI Daisy Configuration for Multiple Peripherals: Questions

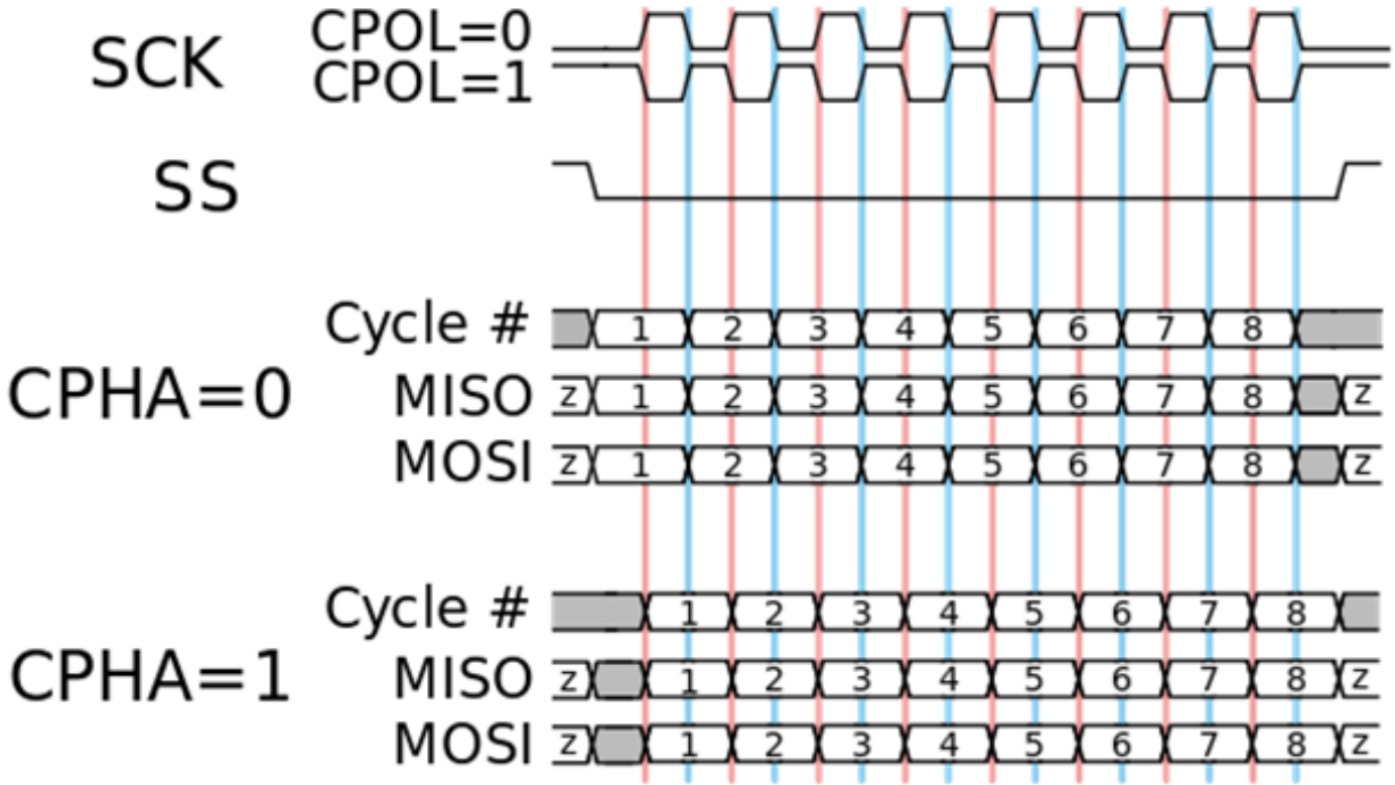
- How many cycles does it take for the data from slave #1 to be received by the master in the daisy-chain configuration?
  - 4, 8, 16, 24 or ???

- After 8 cycles the byte from the shift register of slave #1 ends up in the shift register of slave #2
  - After another 8 cycles, that byte is in the shift register of slave #3
  - After another 8 cycles, the byte is fully copied into the shift register of the master
    - A total of 24 cycles

# Controlling the SPI Timing

- CPOL: Clock (idle) Polarity; CPHA: Clock Phase
  - These timing parameters are configurable
- CPOL determines the base value of the clock
  - 0 – Clock is normally *low*
  - 1 – Clock is normally *high*
- CPHA determines the clock edge on which data is sampled
  - 0 – Sample on the *leading* edge
  - 1 – Sample on the *trailing* edge
- The Master and the Slave must agree on the CPOL and the CPHA
- NOTE: Leading edge may not be the same as the rising edge
- NOTE: Trailing edge may not be the same as the falling edge

# Representative SPI Timing Diagram




- NOTE: Leading edge when CPOL=1 is the falling edge!
- NOTE: Data is stable at the leading edge

Blue: bottom example; Red: top example

# Example: Test you knowledge

- For CPOL $=$ CPHA $= 1$ when is the data stable?
  - CPOL $= 1$: clock normally high
  - CPHA $= 1$: sample on trailing edge

# Example: Test you knowledge

- For $\mathrm{CPOL} = \mathrm{CPHA} = 1$ when is the data stable?
  - CPOL $= 1$: clock normally high
  - CPHA $= 1$: sample on trailing edge

  - Clock high transitions to Clock low
  - Clock falling edge is the leading edge
  - **Clock rising edge is the trailing edge**